

University of Batna 2

Faculty of Mathematics and Computer Science

Departement of Mathematics

Chapter 04: Real functions of a real variable

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$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$$

$$\lim_{x \rightarrow 0} \frac{\log(1 + x)}{x} = 1$$



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I. Limits and continuity

a) Generalities

In this chapter, we are interested in the study of a numerical function of a real variable or simply a function defined from a non-empty part of $E \subset \mathbb{R}$ to $\mathbb{R}$, we note

$$f: E \rightarrow \mathbb{R}$$

$$x \mapsto f(x)$$

f is said:

- a) **Even** if, $\forall x \in E, -x \in E$ and $\forall x \in E, f(-x) = f(x)$
- b) **Odd** if, $\forall x \in E, -x \in E$ and $\forall x \in E, f(-x) = -f(x)$
- c) **Periodic** if, it exists $p \in E$ such that $\forall x \in E, x + p \in E$ and $\forall x \in E, f(x + p) = f(x)$ p is a period of f
- d) The set $\Gamma_f := \{(x, f(x)); x \in E\} \subset \mathbb{R}^2$ is the **graph** of f
- e) **Increasing** in E , if $\forall x, y \in E$ checking $x \leq y$ the $f(x) \leq f(y)$
- f) **Decreasing** in E , if $\forall x, y \in E$ checking $x \leq y$ the $f(x) \geq f(y)$
- g) **Monotonic** if it is **increasing** or **decreasing**
- h) **Upper bounded** if there exists $M \in \mathbb{R}$ such that $f(x) \leq M, \forall x \in \mathbb{R}$
- i) **Lower Bounded** if there exists $m \in \mathbb{R}$ such that $f(x) \geq m, \forall x \in \mathbb{R}$
- j) **Bounded**, if it is **upper bounded** and **lower bounded** in E , or
 $\exists k > 0, |f(x)| \leq k, \forall x \in E$

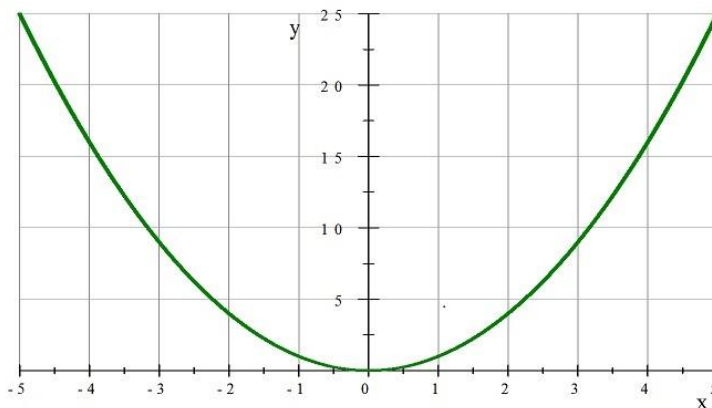
For a bounded function f on E , we set:

$$\sup_E f = \sup f(E) = \sup_{x \in E} f(x) \text{ and } \inf_E f = \inf f(E) = \inf_{x \in E} f(x)$$

And we call them the **upper bound** of f respectively the **lower bound** of f

Examples:

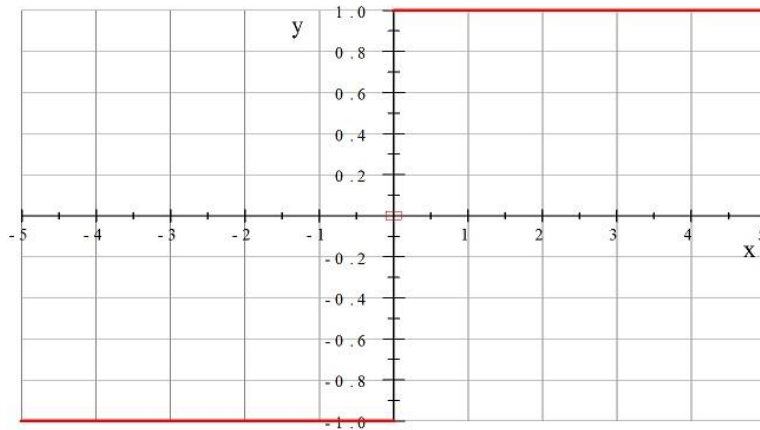
- a) The function $f(x) = x^2$ is defined from $E = \mathbb{R}$ to $\mathbb{R}_+$ and it is even, increasing on $[0, +\infty[$, decreasing on $] -\infty, 0]$ and lower bounded by $0 = \min\{f(x), x \in \mathbb{R}\}$



Graph of the function $f(x) = x^2$

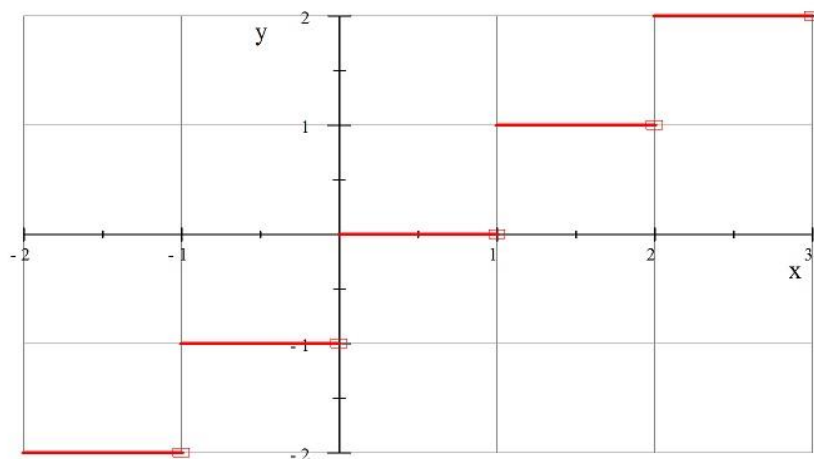
- b) The function $f(n) = n!$ is defined from $E = \mathbb{N}$ to $\mathbb{N}$ is increasing on $\mathbb{N}$ and lower bounded by $1 = \min_{n \in \mathbb{N}} f(n)$.

- c) The function $\text{sign}x = \begin{cases} 1, & \text{if } x > 0 \\ 0, & \text{if } x = 0 \\ -1, & \text{if } x < 0 \end{cases}$ is defined from $\mathbb{R}$ to $\{-1, 0, 1\}$ and it is odd.



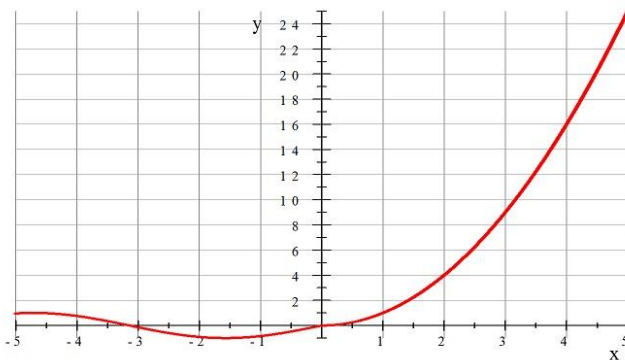
Graph of the function $\text{sign}x$

- d) The function $f(x) = [x]$ is defined from $\mathbb{R}$ in $\mathbb{Z}$ and it is increasing



Graph of the function $f(x) = [x]$

- e) The function $f(x) = \begin{cases} x^2, & \text{if } x > 0 \\ \sin x, & \text{if } x \leq 0 \end{cases}$ is defined from $E = \mathbb{R}$ in $[-1, +\infty[$ and it is periodic on $]-\infty, 0]$ of period $2k\pi, k \in \mathbb{N}^*$



Graph of the function f

b) Limit of a function

We will study the notion of the limit of a sequence of functions defined on a non-empty part of $E \subset \mathbb{R}$.

Definition

Let $x_0 \in \mathbb{R}$ and E a non-empty part of $\mathbb{R}$ and a function f defined from E in $\mathbb{R}$.

We say that f has a limit l at the point x_0 if

$$\forall \varepsilon > 0, \exists \delta > 0: \forall x \in E, x \neq x_0, |x - x_0| < \delta \Rightarrow |f(x) - l| < \varepsilon$$

We write $\lim_{x \rightarrow x_0} f(x) = l$ or $f(x) \rightarrow l$ when $x \rightarrow x_0$.

Examples:

Show that: $\lim_{x \rightarrow 2} (3x + 5) = 11$, $\lim_{x \rightarrow 1} \frac{x}{x+1} = \frac{1}{2}$

Remark

δ is not unique, so we will have to demonstrate the existence of a δ and not calculate all possible δ .

The following theorem allows us to apply the results to sequences

Theorem

Let $x_0, l \in \mathbb{R}$, E a non-empty part of $\mathbb{R}$ and $f: E \rightarrow \mathbb{R}$, the following two proprieties are equivalent:

- 1) $\lim_{x \rightarrow x_0} f(x) = l$
- 2) For all sequence $(x_n) \subset E$ such that $x_n \neq x_0$ and $\lim_{n \rightarrow +\infty} x_n = x_0$, we have $\lim_{n \rightarrow +\infty} f(x_n) = l$.

Examples:

- $f(x) = \frac{2x}{x+1}, x_n = 1 + \frac{1}{n}$
- $f(x) = \sin\left(\frac{1}{x}\right), x \neq 0, x_n = \frac{1}{n\pi}, y_n = \frac{1}{\frac{\pi}{2} + 2n\pi}$

Definition

Let $E \subset \mathbb{R}, E \neq \emptyset$ and $f: E \rightarrow \mathbb{R}, x_0 \in \mathbb{R}$

- 1) We say that f has a **right-hand** limit at the point x_0 if there exists $l \in \mathbb{R}$ such that $\forall \varepsilon > 0, \exists \delta > 0; \forall x \in E$ checking $x_0 < x < x_0 + \delta$ we have $|f(x) - l| < \varepsilon$ and we write $\lim_{x \rightarrow x_0^+} f(x) = l$.
- 2) We say that f has a **left-hand** limit at the point x_0 if there exists $l \in \mathbb{R}$ such that $\forall \varepsilon > 0, \exists \delta > 0; \forall x \in E$ checking $x_0 - \delta < x < x_0$ we have $|f(x) - l| < \varepsilon$ and we write $\lim_{x \rightarrow x_0^-} f(x) = l$.

Remarks

- 1) The limit of a function, when it exists, is unique
- 2) $\lim_{x \rightarrow x_0} f(x) = l \Leftrightarrow \lim_{x \rightarrow x_0^+} f(x) = \lim_{x \rightarrow x_0^-} f(x) = l$

Example:

Calculate $\lim_{x \rightarrow 0} \frac{|x| + x^2}{x}$

Operations on Limits

Definition

Let E be a non-empty subset of $\mathbb{R}$, and let $\mathcal{F}(E, \mathbb{R})$ be the set of functions defined from E into $\mathbb{R}$. Then, $\forall f, g \in \mathcal{F}$

- $(f + g)(x) = f(x) + g(x), \forall x \in E$
- $(fg)(x) = f(x)g(x), \forall x \in E$
- $(\lambda f)(x) = \lambda f(x), \forall x \in E, \forall \lambda \in \mathbb{R}$
- $\left(\frac{1}{g}\right)(x) = \frac{1}{g(x)}, \forall x \in E, g(x) \neq 0,$
- $f \leq g \Leftrightarrow f(x) \leq g(x), \forall x \in E$

Theorem

Let $f, g \in \mathcal{F}(E, \mathbb{R})$ such that $\lim_{x \rightarrow x_0} f(x) = l$ and $\lim_{x \rightarrow x_0} g(x) = l'$, then

- $\lim_{x \rightarrow x_0} [f(x) + g(x)] = l + l'$
- $\lim_{x \rightarrow x_0} [f(x)g(x)] = ll'$
- $\lim_{x \rightarrow x_0} [\lambda f(x)] = \lambda l$
- $\lim_{x \rightarrow x_0} \left[\frac{1}{g(x)}\right] = \frac{1}{l'}, g(x) \neq 0 \text{ and } l' \neq 0$
- If $f(x) < g(x)$ then $l \leq l'$

Infinite limits

Definition

Let $x_0, l \in \mathbb{R}$ and E a non-empty part of $\mathbb{R}$ and $f: E \rightarrow \mathbb{R}$ then

- $\lim_{x \rightarrow +\infty} f(x) = l \Leftrightarrow \forall \varepsilon > 0, \exists A > 0; \forall x \in \mathbb{R}, x > A \Rightarrow |f(x) - l| < \varepsilon$
- $\lim_{x \rightarrow -\infty} f(x) = l \Leftrightarrow \forall \varepsilon > 0, \exists A > 0; \forall x \in \mathbb{R}, x < -A \Rightarrow |f(x) - l| < \varepsilon$
- $\lim_{x \rightarrow x_0} f(x) = +\infty \Leftrightarrow \forall A > 0, \exists \delta > 0; \forall x \in \mathbb{R}, |x - x_0| < \delta \Rightarrow f(x) > A$
- $\lim_{x \rightarrow x_0} f(x) = -\infty \Leftrightarrow \forall A > 0, \exists \delta > 0; \forall x \in \mathbb{R}, |x - x_0| < \delta \Rightarrow f(x) < -A$
- $\lim_{x \rightarrow +\infty} f(x) = +\infty \Leftrightarrow \forall A > 0, \exists B > 0; \forall x \in \mathbb{R}, x > B \Rightarrow f(x) > A$
- $\lim_{x \rightarrow -\infty} f(x) = +\infty \Leftrightarrow \forall A > 0, \exists B > 0; \forall x \in \mathbb{R}, x < -B \Rightarrow f(x) > A$
- $\lim_{x \rightarrow +\infty} f(x) = -\infty \Leftrightarrow \forall A > 0, \exists B > 0; \forall x \in \mathbb{R}, x > B \Rightarrow f(x) < -A$
- $\lim_{x \rightarrow -\infty} f(x) = -\infty \Leftrightarrow \forall A > 0, \exists B > 0; \forall x \in \mathbb{R}, x < -B \Rightarrow f(x) < -A$

Examples: Show that

- $\lim_{x \rightarrow +\infty} \frac{1}{x} = 0$
- $\lim_{x \rightarrow 1} \frac{1}{(1-x)^2} = +\infty$
- $\lim_{x \rightarrow +\infty} x^2 = +\infty$

Remark

In the calculation of a limit of a function at the point x_0 , the forms $\frac{0}{0}, +\infty - \infty, \frac{\infty}{\infty}, 0\infty, 1^\infty, 0^0$ and ∞^0 are considered indeterminate forms. To resolve the indeterminacy is to seek the limit by considering specific transformations.

Example:

Find the following limits:

$$\lim_{x \rightarrow 1} \frac{1-x^2}{1-x^3}, \lim_{x \rightarrow 1^+} \left[\frac{1}{(x-1)^3} - \frac{1}{(x-1)^2} \right], \lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}}, \lim_{x \rightarrow 0} x^x.$$

c) Continuity of a function

Definition

Let E be a non-empty part of $\mathbb{R}$ and $x_0 \in \mathbb{R}$, we say that a function $f: E \rightarrow \mathbb{R}$ is:

- **Continuous** in x_0 if and only if:

$$\forall \varepsilon > 0, \exists \delta > 0: \forall x \in E, x \neq x_0, |x - x_0| < \delta \Rightarrow |f(x) - f(x_0)| < \varepsilon$$

And we write $\lim_{x \rightarrow x_0} f(x) = f(x_0)$.

- **Continuous on the right** at x_0 if and only if:

$$\forall \varepsilon > 0, \exists \delta > 0: \forall x \in E, x \neq x_0, x - x_0 < \delta \Rightarrow |f(x) - f(x_0)| < \varepsilon$$

And we write $\lim_{x \rightarrow x_0^+} f(x) = f(x_0)$.

- **Continuous on the left** at x_0 if and only if:

$$\forall \varepsilon > 0, \exists \delta > 0: \forall x \in E, x \neq x_0, x_0 - x < \delta \Rightarrow |f(x) - f(x_0)| < \varepsilon$$

And we write $\lim_{x \rightarrow x_0^-} f(x) = f(x_0)$.

- **Continues on E** if it is continuous at every point of E

Remark

- $\lim_{x \rightarrow x_0} f(x) = f(x_0) \Leftrightarrow \lim_{x \rightarrow x_0} f(x) = \lim_{x \rightarrow x_0^-} f(x) = f(x_0)$.
- $\lim_{x \rightarrow x_0} f(x) = f(x_0) \Leftrightarrow \forall x_n \in E, x_n \rightarrow x_0$ we have $\lim_{n \rightarrow +\infty} f(x_n) = f(x_0)$.

Examples:

- $f(x) = \begin{cases} \frac{x^2-1}{x-1}, & \text{if } x \neq 1 \\ 3, & \text{if } x = 1 \end{cases}$
- $f(x) = \begin{cases} \frac{\sin x}{x}, & \text{if } x < 0 \\ x^3 + x + 1, & \text{if } x \geq 0 \end{cases}$
- $f(x) = \begin{cases} \frac{e^x-1}{x}, & \text{if } x \neq 0 \\ 1, & \text{if } x = 0 \end{cases}$

Definition (Extension by continuity)

Let E be a non-empty part of $\mathbb{R}$, $x_0 \in \mathbb{R}$ and $f: E \rightarrow \mathbb{R}$ such that $\lim_{x \rightarrow x_0} f(x) = l \in \mathbb{R}$

The function $\check{f}(x) = \begin{cases} f(x), & \text{if } x \neq x_0 \\ l, & \text{if } x = x_0 \end{cases}$

$\check{f}$ is called the **extension by continuity** of f at point x_0

Example:

$f(x) = \frac{x^2 - 5x + 6}{x - 2}$ Can the function f be continuously extended at $x_0 = 2$

$g(x) = \frac{e^x + \sin x - 1}{x}$, Can the function g be continuously extended at $x_0 = 0$

Definition (Uniform continuity)

Let E be a non-empty part of $\mathbb{R}$, we say that $f: E \rightarrow \mathbb{R}$ is **uniformly continuous** on E if and only if,

$$\forall \varepsilon > 0, \exists \delta > 0: \forall x', x'' \in E, |x - x'| < \delta \Rightarrow |f(x) - f(x')| < \varepsilon$$

Remarks

- The number δ depends only on ε in the definition of uniform continuity which is a property on the whole part E .
- Uniform continuity implies continuity on E , indeed, it suffices to take $x' = x$ and $x'' = x_0$.

Example: $f(x) = \cos x$ **Proposition**

Let $E \subset \mathbb{R}, E \neq \emptyset, x_0 \in E$ and $f, g: E \rightarrow \mathbb{R}$ two continuous functions in x_0 , then:
 $f + g, f \cdot g$ and $\frac{1}{g}, g \neq 0$ are continuous in x_0 .

Proposition

Let E, F be two non-empty parts of $\mathbb{R}, x_0 \in E, f: E \rightarrow F$ and $g: F \rightarrow \mathbb{R} (f(E) \subset F)$
 If f is continuous in x_0 and g is continuous in $f(x_0)$ then $g \circ f$ is continuous in x_0 .

d) Theorems on continuous functions on $\mathbb{R}$ **Theorem (Heine)**

Any continuous function on $[a, b]$ is uniformly continuous on this interval.

Theorem (Weierstrass)

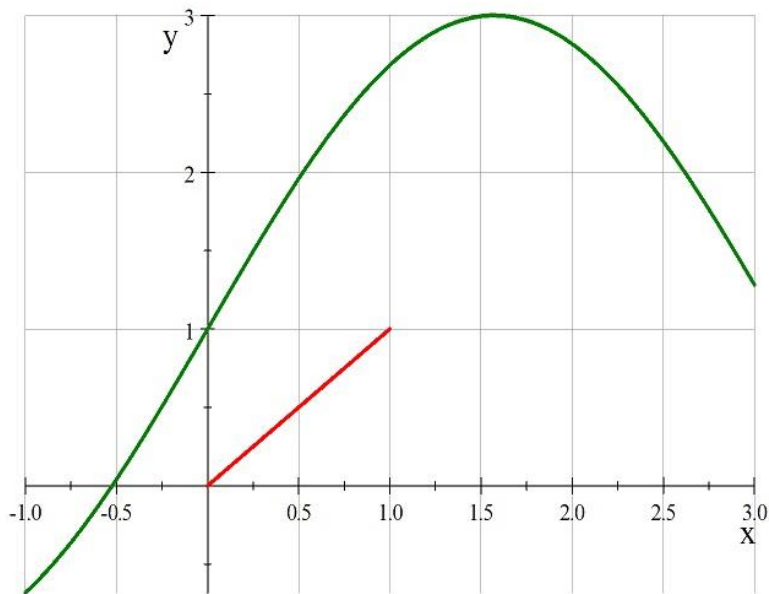
Let f be a continuous function on $[a, b]$, then:

- f is bounded on $[a, b]$
- There exist $x_1, x_2 \in [a, b]$ such that $f(x_1) = \min_{x \in [a, b]} f(x)$ and $f(x_2) = \max_{x \in [a, b]} f(x)$

(f is bounded and attains its bounds on $[a, b]$)

Examples:

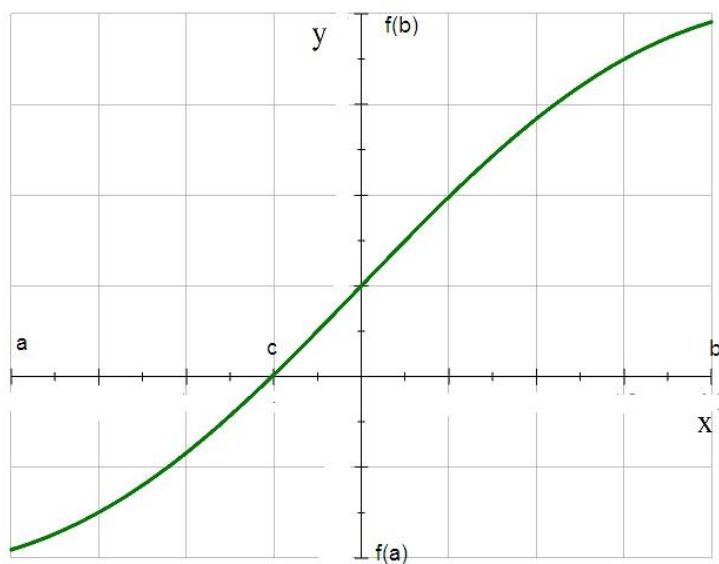
- The function $f(x) = x$ is bounded on $[0, 1]$
- The function $g(x) = 1 + 2 \sin x$ is bounded on $[-1, 3]$



Graphs of the functions f and g

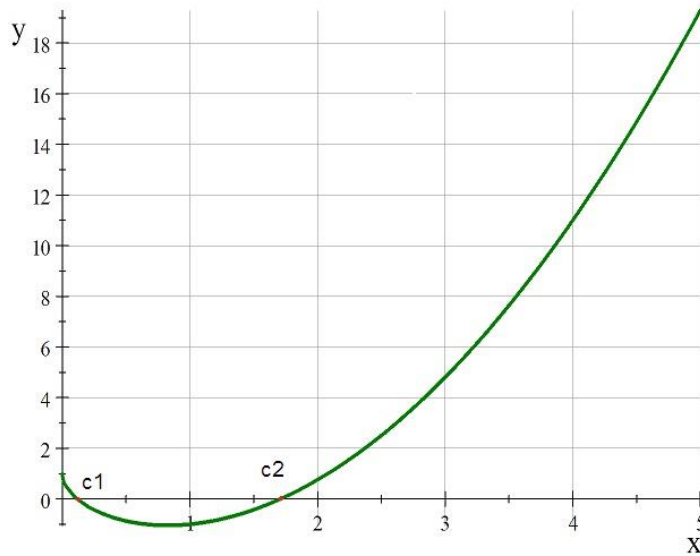
Proposition

Let f be a continuous and bounded function on the interval $[a, b]$ such that $f(a) \cdot f(b) < 0$ then there exists $c \in [a, b]$ such that $f(c) = 0$.

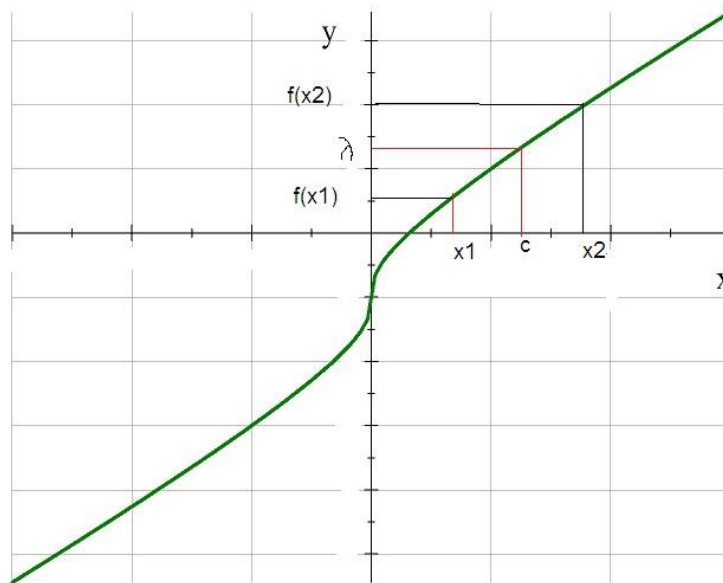


Example:

The equation $x^2 - 3\sqrt{x} + 1 = 0$ has two solutions on $[0, 1]$ and $[1, 2]$

**Theorem (Intermediate value theorem)**

Let f be a function defined and continuous on an interval $I \subset \mathbb{R}$ and $x_1, x_2 \in I$ checking $x_1 < x_2$, then for all $f(x_1) < \lambda < f(x_2)$ there exists $c \in]x_1, x_2[$ such that $f(c) = \lambda$



Proposition

Let I be an interval and $f: I \rightarrow \mathbb{R}$ a continuous function then $f(I)$ is an interval.

Remarks

- If f is a continuous on $[a, b]$ then $f[a, b] = [m, M]$ where $m = \min_{x \in [a, b]} f(x)$ and $M = \max_{x \in [a, b]} f(x)$.
- In the case where f is continuous on a non-closed interval I , $f(I)$ is not necessarily non-closed.

Examples:

- The function $f(x) = x^2$ is continuous on the open interval $] -1, 1[$, but $f(] -1, 1[) = [0, 1[$
- The function $f(x) = \sin x$ is continuous on the open interval $] -\pi, \pi[$, but $f(] -\pi, \pi[) = [-1, 1]$.

e) Monotonic function and continuity

Note that if I is any interval of $\mathbb{R}$ and $f: I \rightarrow \mathbb{R}$ is a strictly monotone function, then if $f(I)$ is an interval, the reciprocal (or inverse) function f^{-1} of f is defined on $f(I)$ and is also strictly monotone.

f^{-1} is strictly increasing if f is strictly increasing.

f^{-1} is strictly decreasing if f is strictly decreasing.

Indeed, $\forall x \in I$ and $\forall y \in f(I)$ we have $y = f(f^{-1}(y))$ and $x = f^{-1}(y)$ et then

$\forall y_1, y_2 \in f(I)$ such that $y_1 < y_2$ we have

$$\frac{f^{-1}(y_1) - f^{-1}(y_2)}{y_1 - y_2} = \frac{x_1 - x_2}{f(x_1) - f(x_2)}.$$

Proposition

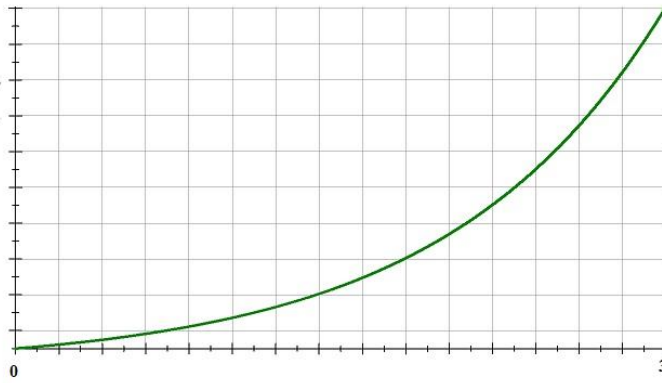
let I be any interval of $\mathbb{R}$ and $f: I \rightarrow \mathbb{R}$ a strictly monotone function, then f is continuous on I if and only if $f(I)$ is an interval.

Theorem (Inverse Function Theorem)

Let I an interval and $f: I \rightarrow \mathbb{R}$ be a continuous and strictly monotonic function, then, the reciprocal function $f^{-1}: f(I) \rightarrow I$ exists and it is continuous.

Proposition

Let I an interval and $f: I \rightarrow \mathbb{R}$ be a continuous function, then, f is injective if and only if f is strictly monotonic.



f) Landau notation

The **Landau notation**, also known as asymptotic notation, is a mathematical notation used to describe the limiting behavior of functions as their argument tends to a particular value, typically infinity. It's widely used in fields like mathematics, computer science, and complexity theory to describe the growth rates of functions, especially when analyzing the performance of algorithms.

Types of Landau Notation

➤ Big O Notation ($O(f(x))$): Upper Bound

- $f(x) = O(g(x))$ means that $f(x)$ grows at most as fast as $g(x)$ when x becomes large.
- **Formally:** $f(x) = O(g(x))$ if there exist constants $C > 0$ and x_0 such that $|f(x)| \leq C |g(x)|$ for all $x \geq x_0$.

➤ Big Omega Notation ($\Omega(f(x))$): Lower Bound

- $f(x) = \Omega(g(x))$ means that $f(x)$ grows at least as fast as $g(x)$ for large x .
- **Formally:** $f(x) = \Omega(g(x))$ if there exist constants $C > 0$ and x_0 such that $f(x) \geq Cg(x)$ for all $x \geq x_0$.

➤ Big Theta Notation ($\Theta(f(x))$): Tight Bound

- $f(x) = \Theta(g(x))$ means that $f(x)$ grows at the same rate as $g(x)$ for large x .
- **Formally:** $f(x) = \Theta(g(x))$ if there exist constants $C_1 > 0, C_2 > 0$ and x_0 such that for all $x \geq x_0$, $C_1g(x) \leq f(x) \leq C_2g(x)$.

➤ Little o Notation ($o(f(x))$): Strict Upper Bound

- $f(x) = o(g(x))$ means that $f(x)$ grows strictly slower than $g(x)$ as $x \rightarrow \infty$.
- **Formally:** $f(x) = o(g(x))$ if for every constant $\epsilon > 0$, there exists an x_0 such that $|f(x)| < \epsilon |g(x)|$ for all $x > x_0$.

➤ Little Omega Notation ($\omega(f(x))$): Strict Lower Bound

- $f(x) = \omega(g(x))$ means that $f(x)$ grows strictly faster than $g(x)$ as $x \rightarrow \infty$.
- **Formally:** $f(x) = \omega(g(x))$ if for every constant $\epsilon > 0$, there exists an x_0 such that $|f(x)| > \epsilon |g(x)|$ for all $x > x_0$.